
SMART AGRI-TOGO

Complete Technical Course

From Mathematics to the Living Farm

Differential Equations · Optimal Control · Machine Learning
Soil Physics · Irrigation Science · Organic Fertilization
Biogas & Composting · Solar Energy Systems

With worked examples and exercises for self-study

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A living document — to be updated as the project progresses

Preface

This course is your personal companion for the Smart Agri-Togo project. It is not a textbook written for a general audience — it is written specifically *for you*, starting from your background in theoretical physics and building up every mathematical, agronomic, and engineering concept you will need to design, build, and operate an intelligent counter-season farm in Togo.

The course is organised in four parts:

- **Part I — Mathematics:** the differential equations, linear algebra, optimisation theory, optimal control, and machine learning that power the intelligent irrigation system.
- **Part II — Irrigation Science:** soil physics, evapotranspiration, and crop water models — translating your mathematical tools into real agronomic quantities.
- **Part III — Organic Fertilization:** the complete plan to nourish your crops using only biomass, animal manure, and human biosolids — no synthetic chemicals — with the biochemistry and the practical protocols.
- **Part IV — Energy Systems:** a rigorous analysis of your energy needs and a full solar power system design, sized for your specific conditions in southern Togo.

Every chapter contains **worked examples** drawn from the actual Smart Agri-Togo context and **exercises with full solutions**. The exercises are not optional — they are the fastest path to genuine understanding.

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Part I

Mathematical Foundations

Chapter 1

Ordinary Differential Equations and Dynamical Systems

1.1 Why ODEs? The Physics of Soil Water

At the heart of Smart Agri-Togo lies one equation: the soil water balance. Like all physical conservation laws, it is a differential equation. Understanding it mathematically is the first step to controlling it optimally.

Definition: Ordinary Differential Equation (ODE)

An **ordinary differential equation** of order n for the unknown function $x(t)$ is a relation of the form

$$F(t, x, x', x'', \dots, x^{(n)}) = 0.$$

It is called *autonomous* if t does not appear explicitly. It is called *linear* if it is linear in x and all its derivatives.

1.2 The Soil Water Balance ODE

Let $\theta(t)$ [m³/m³] be the volumetric soil water content in a single field cell, averaged over root depth z_r [m]. The mass balance over a unit surface area gives:

Soil Water Balance (single cell)

$$z_r \frac{d\theta}{dt} = I(t) + P(t) - ET_c(t) - D(t) \quad (1.1)$$

where I [mm/h] is the irrigation rate (the *control input*), P [mm/h] is precipitation, ET_c [mm/h] is crop evapotranspiration, and D [mm/h] is drainage below

the root zone.

In the dry season, $P \approx 0$ and $D \approx 0$ when $\theta < \theta_{fc}$. Equation (1.1) reduces to:

$$z_r \frac{d\theta}{dt} = I(t) - ET_c(t), \quad (1.2)$$

which is a *first-order linear non-autonomous ODE* (non-autonomous because I and ET_c are functions of time).

1.2.1 General Solution by Integration

Rewrite Eq. (1.2) as:

$$\frac{d\theta}{dt} = f(t), \quad f(t) = \frac{I(t) - ET_c(t)}{z_r}. \quad (1.3)$$

By direct integration, starting from initial condition $\theta(0) = \theta_0$:

$$\theta(t) = \theta_0 + \int_0^t f(\tau) d\tau = \theta_0 + \frac{1}{z_r} \int_0^t [I(\tau) - ET_c(\tau)] d\tau. \quad (1.4)$$

This tells us directly how the moisture changes due to net water input.

1.3 Numerical Solution: Euler and Runge-Kutta

In practice, $I(t)$ and $ET_c(t)$ are only known at discrete time steps (e.g., every 30 minutes). We use numerical integration.

1.3.1 Euler Method

With time step Δt , the Euler method discretises Eq. (1.2) as:

$$\theta^{k+1} = \theta^k + \Delta t \cdot f(t_k) = \theta^k + \frac{\Delta t}{z_r} [I^k - ET_c^k]. \quad (1.5)$$

This is the equation the Raspberry Pi solves at every time step. It is first-order accurate: error $\propto (\Delta t)^2$ per step.

1.3.2 Runge-Kutta 4th Order (RK4)

For systems where higher accuracy matters (e.g., calibrating soil models), RK4 uses four evaluations of f :

$$k_1 = f(t_n, \theta^n), \quad (1.6)$$

$$k_2 = f\left(t_n + \frac{\Delta t}{2}, \theta^n + \frac{\Delta t}{2} k_1\right), \quad (1.7)$$

$$k_3 = f\left(t_n + \frac{\Delta t}{2}, \theta^n + \frac{\Delta t}{2} k_2\right), \quad (1.8)$$

$$k_4 = f(t_n + \Delta t, \theta^n + \Delta t k_3), \quad (1.9)$$

$$\theta^{n+1} = \theta^n + \frac{\Delta t}{6}(k_1 + 2k_2 + 2k_3 + k_4). \quad (1.10)$$

RK4 is fourth-order accurate (error $\propto (\Delta t)^5$ per step). For the soil water balance, the Euler method with $\Delta t = 30$ min is sufficient, but RK4 is used when calibrating the van Genuchten soil model.

Example — Daily moisture dynamics for onion (dry season)

Given: $z_r = 0.3$ m, $\theta_0 = 0.60$, $ET_c = 0.3$ mm/h (constant), $I(t) = 0.8$ mm/h for $t \in [6 \text{ h}, 9 \text{ h}]$ and $I = 0$ otherwise. $\Delta t = 1$ h. Find $\theta(t)$ for $t = 0$ to $t = 24$ h using Euler.

Solution: $f(t) = (I - ET_c)/z_r$ so $f = (0 - 0.3)/0.3 = -1$ mm/(h·m³/m³) when not irrigating, and $f = (0.8 - 0.3)/0.3 = 1.667$ when irrigating.

Euler steps ($\Delta t = 1$ h): $t = 0$: $\theta^1 = 0.60 + 1 \times (-1) \times 10^{-3} = 0.600 - 0.001 = 0.599$. At $t = 6$ h, $\theta = 0.594$. During irrigation ($t = 6, 7, 8$): θ increases by $1.667 \times 10^{-3} \approx 0.001667$ per hour. After 3 h irrigation: $\theta(9) \approx 0.594 + 3 \times 0.001667 = 0.599$. Then declines again at -0.001 /h until $t = 24$: $\theta(24) \approx 0.599 - 15 \times 0.001 = 0.584$.

Exercise 1.1.1

A carrot cell has $z_r = 0.25$ m, $\theta_0 = 0.45$, and $ET_c = 0.25$ mm/h. Irrigation is applied at rate $I = 1.0$ mm/h for 4 hours starting at $t = 8$ h. Using the Euler method with $\Delta t = 1$ h:

- Compute $\theta(t)$ from $t = 0$ to $t = 24$ h.
- At what time does θ first drop below the wilting-point threshold $\theta_{wp} = 0.15$?
- Plot (sketch) $\theta(t)$.

Solution to Exercise 1.1.1

(a) $f_{dry} = (0 - 0.25)/0.25 = -1.0$ (mm/h)/(m³/m³) = -0.001 per hour in practice since I and ET_c are in mm/h and z_r is in m (converting: $\Delta\theta = \Delta t \times$

$(I - ET_c)/(z_r \times 1000)$; or keeping all in consistent mm units: $z_r = 250$ mm, so $f_{dry} = -0.25/250 = -0.001 \text{ h}^{-1}$).

Let us keep consistent units throughout (all lengths in mm, $z_r = 250$ mm):

$$\theta^{k+1} = \theta^k + \frac{\Delta t}{250}(I^k - ET_c^k)$$

- $t = 0 \rightarrow 8$ h: $I = 0$, $f = -0.25/250 = -0.001$. After 8 h: $\theta(8) = 0.45 - 8 \times 0.001 = 0.442$.
- $t = 8 \rightarrow 12$ h: $I = 1.0$ mm/h, $f = (1.0 - 0.25)/250 = 0.003$. After 4 h: $\theta(12) = 0.442 + 4 \times 0.003 = 0.454$.
- $t = 12 \rightarrow 24$ h: $I = 0$, $f = -0.001$. After 12 h: $\theta(24) = 0.454 - 12 \times 0.001 = 0.442$.

(b) Without irrigation, θ would decline from 0.45 at rate 0.001/h. Time to reach 0.15: $(0.45 - 0.15)/0.001 = 300 \text{ h} \approx 12.5$ days. With irrigation every few days, it never reaches θ_{wp} during the growing season.

(c) The plot shows a piecewise linear decrease, a brief rise during the 4-hour irrigation window ($t = 8$ to 12 h), then a linear decrease again. The shape is a sawtooth wave repeating each irrigation cycle.

1.4 Stability of the Soil Water ODE

Setting $I = ET_c$ (perfectly balanced irrigation), the system reaches a stationary point $\bar{\theta}$. Perturbing: $\theta = \bar{\theta} + \varepsilon$. The linearised equation is $\dot{\varepsilon} = -\lambda\varepsilon$ where $\lambda > 0$ (because ET_c increases with θ through the stress coefficient K_s). This means the equilibrium is **stable**: small perturbations decay exponentially.

Chapter 2

Linear Algebra for Multi-Cell Systems

2.1 From One Cell to 25 Cells

The 5×5 field has $N = 25$ cells, each with its own moisture θ_i . We group them into a **state vector**:

$$\boldsymbol{\theta} = (\theta_1, \theta_2, \dots, \theta_{25})^T \in \mathbb{R}^{25}. \quad (2.1)$$

Similarly, the irrigation rates form a **control vector** $\mathbf{u} = (I_1, \dots, I_{25})^T \in \mathbb{R}^{25}$.

2.2 State-Space Representation

Writing Eq. (1.5) for all cells simultaneously:

$$\boldsymbol{\theta}^{k+1} = A\boldsymbol{\theta}^k + B\mathbf{u}^k + \mathbf{d}^k, \quad (2.2)$$

where:

- $A = I_{25} - \frac{\Delta t}{z_r} \text{diag}(\lambda_1, \dots, \lambda_{25})$ is the state transition matrix (I_{25} is the identity), with λ_i accounting for drainage and stress-dependent ET.
- $B = \frac{\Delta t}{z_r} I_{25}$ is the input matrix (diagonal, since each valve controls only its own cell).
- $\mathbf{d}^k = \frac{\Delta t}{z_r} (-ET_{c,1}^k, \dots, -ET_{c,25}^k)^T$ is the disturbance vector.

In the simplified model (no lateral flow), A and B are both **diagonal** — the 25-cell system decouples into 25 independent scalar ODEs.

2.3 Key Matrix Operations

Definition: Matrix-Vector Product

If $A \in \mathbb{R}^{m \times n}$ and $\mathbf{x} \in \mathbb{R}^n$, then $(A\mathbf{x})_i = \sum_{j=1}^n A_{ij}x_j$.

Definition: Eigenvalues and Eigenvectors

$\lambda \in \mathbb{R}$ is an eigenvalue of A with eigenvector $\mathbf{v} \neq \mathbf{0}$ if $A\mathbf{v} = \lambda\mathbf{v}$. The eigenvalues determine system stability: if all $|\lambda_i| < 1$ (discrete time) or $\text{Re}(\lambda_i) < 0$ (continuous time), the system is stable.

For our diagonal system, eigenvalues are the diagonal entries A_{ii} . Since $A_{ii} = 1 - \frac{\Delta t \lambda_i}{z_r} < 1$ for $\lambda_i > 0$ and small Δt , the system is stable.

2.4 Lifted (Stacked) Representation for MPC

Model Predictive Control requires predicting the future state trajectory over a horizon H steps. Stacking states and inputs:

$$\mathbf{X} = \begin{pmatrix} \boldsymbol{\theta}^1 \\ \boldsymbol{\theta}^2 \\ \vdots \\ \boldsymbol{\theta}^H \end{pmatrix} \in \mathbb{R}^{NH}, \quad \mathbf{U} = \begin{pmatrix} \mathbf{u}^0 \\ \mathbf{u}^1 \\ \vdots \\ \mathbf{u}^{H-1} \end{pmatrix} \in \mathbb{R}^{NH},$$

the prediction is compactly written as:

$$\mathbf{X} = \mathcal{A}\boldsymbol{\theta}^0 + \mathcal{B}\mathbf{U} + \mathcal{D}, \quad (2.3)$$

where $\mathcal{A} \in \mathbb{R}^{NH \times N}$, $\mathcal{B} \in \mathbb{R}^{NH \times NH}$ (lower triangular block Toeplitz matrix), and $\mathcal{D}$ collects the disturbances. This **lifted system** converts the MPC problem into a single quadratic program (Chapter 4).

Example — 3-cell lifted system

Let $N = 3$ cells, $H = 2$ steps, $A = 0.98 I_3$, $B = 0.01 I_3$.

$$\mathcal{A} = \begin{pmatrix} A \\ A^2 \end{pmatrix} = \begin{pmatrix} 0.98 & 0 & 0 \\ 0 & 0.98 & 0 \\ 0 & 0 & 0.98 \\ 0.9604 & 0 & 0 \\ 0 & 0.9604 & 0 \\ 0 & 0 & 0.9604 \end{pmatrix},$$

$$\mathcal{B} = \begin{pmatrix} B & 0 \\ AB & B \end{pmatrix} = \begin{pmatrix} 0.01I_3 & 0 \\ 0.0098I_3 & 0.01I_3 \end{pmatrix}.$$

Given $\boldsymbol{\theta}^0 = (0.5, 0.4, 0.6)^T$ and $\mathbf{U} = (I_3, I_3)^T \cdot 2$ (irrigation rate 2 mm/h in all cells):

$$\boldsymbol{\theta}^1 = 0.98\boldsymbol{\theta}^0 + 0.01 \times 2\mathbf{1}_3 = (0.49 + 0.02, 0.392 + 0.02, 0.588 + 0.02) = (0.51, 0.412, 0.608).$$

Exercise 2.2.1

A simplified 4-cell system has $z_r = 300$ mm, $\Delta t = 0.5$ h, $ET_c = [0.3, 0.25, 0.4, 0.2]^T$ mm/h, $I = [0.8, 0, 0.8, 0]^T$ mm/h.

- Write A and B explicitly.
- Starting from $\boldsymbol{\theta}^0 = (0.55, 0.60, 0.48, 0.65)^T$, compute $\boldsymbol{\theta}^1$ and $\boldsymbol{\theta}^2$.
- Which cells need irrigation most urgently if $\theta_{low} = 0.40$?

Solution to Exercise 2.2.1

(a) $B = \frac{\Delta t}{z_r} I_4 = \frac{0.5}{300} I_4$. For A : since we include a simple ET loss term, $A = (1 - \frac{\Delta t ET_c}{z_r \theta}) I$ but in the linearised version around an operating point $A \approx I_4$ (drainage negligible, stress $K_s \approx 1$). More precisely: $d^k = -\frac{\Delta t}{z_r} ET_c$. So $A = I_4$, $B = \frac{1}{600} I_4$, $d^k = \frac{-0.5}{300} [0.3, 0.25, 0.4, 0.2]^T = [-0.0005, -0.000417, -0.000667, -0.000333]^T$.

(b) $\boldsymbol{\theta}^1 = \boldsymbol{\theta}^0 + B\mathbf{u}^0 + d^0$:

$$\theta_1^1 = 0.55 + \frac{0.8}{600} - 0.0005 = 0.55 + 0.001333 - 0.0005 = 0.5508$$

$$\theta_2^1 = 0.60 + 0 - 0.000417 = 0.5996$$

$$\theta_3^1 = 0.48 + 0.001333 - 0.000667 = 0.4807$$

$$\theta_4^1 = 0.65 + 0 - 0.000333 = 0.6497$$

Step 2: repeat with same inputs $\rightarrow \theta_3$ continues to be the lowest.

(c) No cell is below 0.40 yet. Cell 3 (moisture 0.4807, declining most rapidly at rate 0.000667/step) will reach 0.40 fastest: time = $(0.4807 - 0.40)/0.000667 \approx 121$ steps ≈ 60.5 h without irrigation.

Chapter 3

Optimisation Theory

3.1 The Core Problem

All intelligent decisions in Smart Agri-Togo reduce to: “find the irrigation strategy $\mathbf{u}$ that minimises water use while maximising crop yield.” This is an **optimisation problem**.

Definition: Optimisation Problem (general form)

$$\min_{\mathbf{x} \in \mathbb{R}^n} f(\mathbf{x}) \quad \text{subject to} \quad g_i(\mathbf{x}) \leq 0, \quad h_j(\mathbf{x}) = 0.$$

$f : \mathbb{R}^n \rightarrow \mathbb{R}$ is the *objective function*; g_i are inequality constraints; h_j are equality constraints.

3.2 Unconstrained Optimisation

3.2.1 Gradient and Necessary Conditions

For a differentiable $f : \mathbb{R}^n \rightarrow \mathbb{R}$, the gradient is:

$$\nabla f(\mathbf{x}) = \left(\frac{\partial f}{\partial x_1}, \dots, \frac{\partial f}{\partial x_n} \right)^T.$$

A necessary condition for $\mathbf{x}^*$ to be a local minimum: $\nabla f(\mathbf{x}^*) = \mathbf{0}$.

The **Hessian** $H = \nabla^2 f$ is the matrix of second derivatives: $H_{ij} = \frac{\partial^2 f}{\partial x_i \partial x_j}$. If $H(\mathbf{x}^*) \succ 0$ (positive definite), then $\mathbf{x}^*$ is a strict local minimum.

3.2.2 Gradient Descent

Update rule: $\mathbf{x}^{k+1} = \mathbf{x}^k - \alpha \nabla f(\mathbf{x}^k)$, where $\alpha > 0$ is the learning rate. Converges to a local minimum for smooth f with appropriate α .

3.3 Constrained Optimisation: KKT Conditions

Karush-Kuhn-Tucker (KKT) Conditions

For the problem $\min f(\mathbf{x})$ s.t. $g_i(\mathbf{x}) \leq 0$ ($i = 1, \dots, m$), $h_j(\mathbf{x}) = 0$ ($j = 1, \dots, p$), define the **Lagrangian**:

$$L(\mathbf{x}, \boldsymbol{\mu}, \boldsymbol{\lambda}) = f(\mathbf{x}) + \sum_i \mu_i g_i(\mathbf{x}) + \sum_j \lambda_j h_j(\mathbf{x}).$$

At a local minimum $\mathbf{x}^*$, under constraint qualifications:

$$\nabla_{\mathbf{x}} L = \mathbf{0} \quad (\text{stationarity}) \tag{3.1}$$

$$g_i(\mathbf{x}^*) \leq 0, \quad h_j(\mathbf{x}^*) = 0 \quad (\text{primal feasibility}) \tag{3.2}$$

$$\mu_i \geq 0 \quad (\text{dual feasibility}) \tag{3.3}$$

$$\mu_i g_i(\mathbf{x}^*) = 0 \quad \forall i \quad (\text{complementary slackness}) \tag{3.4}$$

μ_i and λ_j are called *Lagrange multipliers* (or dual variables).

3.4 Quadratic Programs (QP)

The MPC irrigation problem is a **Quadratic Program**: objective quadratic, constraints linear:

Standard QP Form

$$\min_{\mathbf{u} \in \mathbb{R}^n} \frac{1}{2} \mathbf{u}^T Q \mathbf{u} + c^T \mathbf{u} \quad \text{s.t.} \quad G \mathbf{u} \leq \mathbf{h}, \quad A_{eq} \mathbf{u} = \mathbf{b}_{eq} \tag{3.5}$$

where $Q \succ 0$ ensures a unique global minimum. Solvable in polynomial time by interior-point or active-set methods. The Python library `cvxpy` calls the solver OSQP to solve this automatically.

Example — Simple 2-cell irrigation QP

Problem: Two cells, one time step. $\theta_1^* = \theta_2^* = 0.65$, current $\theta = (0.40, 0.55)$. Minimise $\frac{1}{2}(\theta_1^1 - 0.65)^2 + \frac{1}{2}(\theta_2^1 - 0.65)^2 + 0.01(I_1^2 + I_2^2)$ subject to $0 \leq I_i \leq 5$ and

$I_1 + I_2 \leq 6$. With $\Delta\theta_i = 0.002 I_i$ (linearised), the QP variables are $I_1, I_2 \geq 0$.

Unconstrained solution: $\frac{d}{dI_1} [0.5(0.40 + 0.002I_1 - 0.65)^2 + 0.01I_1^2] = 0$: $(0.40 + 0.002I_1 - 0.65) \cdot 0.002 + 0.02I_1 = 0 \Rightarrow -0.0005 + 0.000004I_1 + 0.02I_1 = 0 \Rightarrow I_1^* = 0.0005/0.020004 \approx 0.025$ mm/h. Very small because the water-penalty $\beta = 0.01$ is large. With $\beta = 0.001$: $I_1^* \approx 0.247$ mm/h. This shows the role of the weighting parameter β .

Exercise 3.3.1

Write the KKT conditions for the single-cell irrigation problem:

$$\min_{I \geq 0} \frac{\alpha}{2}(\theta_0 + \delta I - \theta^*)^2 + \frac{\beta}{2}I^2$$

where $\delta = \Delta t/z_r$, $\alpha, \beta > 0$. Solve for the optimal I^* as a function of the parameters. Show that: (a) if $\theta_0 \geq \theta^*$, then $I^* = 0$; (b) as $\beta \rightarrow 0$, $I^* \rightarrow (\theta^* - \theta_0)/\delta$ (the exact amount needed to reach target).

Solution to Exercise 3.3.1

The Lagrangian is $L(I, \mu) = \frac{\alpha}{2}(\theta_0 + \delta I - \theta^*)^2 + \frac{\beta}{2}I^2 - \mu I$. KKT conditions: (1) $\frac{\partial L}{\partial I} = \alpha\delta(\theta_0 + \delta I - \theta^*) + \beta I - \mu = 0$; (2) $\mu \geq 0$; (3) $\mu I = 0$; (4) $I \geq 0$.

Case 1: $\mu = 0$ (unconstrained interior solution). $\alpha\delta(\theta_0 + \delta I - \theta^*) + \beta I = 0 \Rightarrow I^* = \frac{-\alpha\delta(\theta_0 - \theta^*)}{\alpha\delta^2 + \beta} = \frac{\alpha\delta(\theta^* - \theta_0)}{\alpha\delta^2 + \beta}$. This is valid (i.e., $I^* \geq 0$) iff $\theta_0 \leq \theta^*$. **(a)** If $\theta_0 \geq \theta^*$, then $I^* \leq 0$ which violates $I \geq 0$, so the constraint is active: $I^* = 0$. **(b)** As $\beta \rightarrow 0$: $I^* \rightarrow \frac{\alpha\delta(\theta^* - \theta_0)}{\alpha\delta^2} = \frac{\theta^* - \theta_0}{\delta}$. QED.

Chapter 4

Optimal Control and Model Predictive Control

4.1 What is Optimal Control?

Optimal control asks: *given a dynamical system, what input signal $u(t)$ minimises a cost J over a time horizon?*

Definition: Finite-Horizon Optimal Control Problem

$$\min_{\mathbf{u}(\cdot)} J = \int_0^{T_f} L(\mathbf{x}, \mathbf{u}, t) dt + \Phi(\mathbf{x}(T_f)) \quad (4.1)$$

$$\text{s.t. } \dot{\mathbf{x}} = f(\mathbf{x}, \mathbf{u}, t), \quad \mathbf{x}(0) = \mathbf{x}_0, \quad (4.2)$$

$$\mathbf{u}(t) \in \mathcal{U}, \quad \mathbf{x}(t) \in \mathcal{X}. \quad (4.3)$$

L is the *running cost*, Φ is the *terminal cost*, $\mathcal{U}$ and $\mathcal{X}$ are the feasible sets for control and state.

4.2 Pontryagin's Maximum Principle (PMP)

For continuous-time problems, the PMP provides necessary conditions via the **Hamiltonian**:

$$H(\mathbf{x}, \mathbf{u}, \mathbf{p}, t) = L(\mathbf{x}, \mathbf{u}, t) + \mathbf{p}(t)^T f(\mathbf{x}, \mathbf{u}, t), \quad (4.4)$$

where $\mathbf{p}(t) \in \mathbb{R}^n$ is the **costate** (or adjoint variable), solving:

$$\dot{\mathbf{p}} = -\frac{\partial H}{\partial \mathbf{x}}, \quad \mathbf{p}(T_f) = \frac{\partial \Phi}{\partial \mathbf{x}}(T_f) \quad (\text{transversality}). \quad (4.5)$$

The optimal control minimises H at each instant:

$$\mathbf{u}^*(t) = \arg \min_{\mathbf{u} \in \mathcal{U}} H(\mathbf{x}^*, \mathbf{u}, \mathbf{p}, t). \quad (4.6)$$

Example — PMP for single-cell irrigation

Cost: $J = \int_0^T [\alpha(\theta - \theta^*)^2 + \beta I^2] dt$. Dynamics: $\dot{\theta} = (I - ET_c)/z_r$. Hamiltonian: $H = \alpha(\theta - \theta^*)^2 + \beta I^2 + p(I - ET_c)/z_r$. Costate: $\dot{p} = -\partial H/\partial \theta = -2\alpha(\theta - \theta^*)$, $p(T) = 0$. Optimal control: $\partial H/\partial I = 2\beta I + p/z_r = 0 \Rightarrow I^*(t) = -p(t)/(2\beta z_r)$. The costate $p(t)$ acts as a “water price”: when $\theta < \theta^*$ (deficit), $p > 0$, pushing $I^* > 0$.

4.3 Dynamic Programming and the Bellman Equation

For discrete-time systems, the optimal cost-to-go $V^*(k, \boldsymbol{\theta})$ satisfies:

Bellman’s Principle of Optimality

$$V^*(k, \boldsymbol{\theta}) = \min_{\mathbf{u}^k \in \mathcal{U}} [L(\boldsymbol{\theta}, \mathbf{u}^k) + V^*(k+1, A\boldsymbol{\theta} + B\mathbf{u}^k)], \quad (4.7)$$

with terminal condition $V^*(H, \boldsymbol{\theta}) = \Phi(\boldsymbol{\theta})$.

The optimal policy $\mathbf{u}^{*k} = \pi^*(k, \boldsymbol{\theta})$ is obtained by solving the Bellman equation backwards from $t = T_f$.

4.4 Model Predictive Control (MPC): Principle and Algorithm

MPC is the **practical implementation** of optimal control in real time. Rather than solving the Bellman equation over the full season (~ 4800 steps), it solves a short-horizon QP at each step and only applies the first control action.

MPC Algorithm (one iteration)

1. **Measure** the current state $\boldsymbol{\theta}^k$ (read all soil moisture sensors).
2. **Forecast** disturbances $\hat{ET}_c^{k:k+H-1}$, $\hat{P}^{k:k+H-1}$ (LSTM output).

3. **Solve** the QP over horizon H :

$$\min_{\mathbf{U}} J = \sum_{j=0}^{H-1} [\alpha \|\boldsymbol{\theta}^{k+j} - \boldsymbol{\theta}^*\|^2 + \beta \|\mathbf{u}^{k+j}\|^2] + \Phi(\boldsymbol{\theta}^{k+H}) \quad (4.8)$$

$$\text{s.t. } \boldsymbol{\theta}^{k+j+1} = A\boldsymbol{\theta}^{k+j} + B\mathbf{u}^{k+j} + \mathbf{d}^{k+j} \quad (4.9)$$

$$\mathbf{0} \leq \mathbf{u}^{k+j} \leq I_{\max} \mathbf{1} \quad (4.10)$$

$$\mathbf{1}^T \mathbf{u}^{k+j} \leq Q_{\max} \quad (4.11)$$

4. **Apply** $\mathbf{u}^k = \mathbf{U}^*(0)$ (first element of optimal sequence).

5. **Advance** $k \leftarrow k + 1$, go to step 1.

The key parameters are:

- H : prediction horizon (we use 48 steps = 24 h with $\Delta t = 30$ min).
- α : crop-stress weight — larger α keeps moisture close to target.
- β : water-use weight — larger β saves water but tolerates more stress.

4.5 PID Control as a Simpler Baseline

The Proportional-Integral-Derivative (PID) controller for irrigation:

$$I(t) = K_p e(t) + K_i \int_0^t e(\tau) d\tau + K_d \dot{e}(t), \quad e(t) = \theta^* - \theta(t). \quad (4.12)$$

The PID is simpler to implement and needs no model or forecast. Its main limitation: it cannot handle constraints ($I \geq 0$, pump capacity) or coordinate multiple valves.

Exercise 4.4.1

Write out the full MPC QP in matrix form (using the lifted system from Eq. (2.3)) for $N = 1$ cell, $H = 3$ steps, $\alpha = 1$, $\beta = 0.1$, $\theta^* = 0.65$, $I_{\max} = 2$ mm/h. Show that it reduces to a 3×3 QP with explicit Q_{QP} and c_{QP} .

Solution to Exercise 4.4.1

With $N = 1$, $H = 3$: $\mathbf{X} = (\theta^1, \theta^2, \theta^3)^T$, $\mathbf{U} = (I^0, I^1, I^2)^T$. $\mathcal{B} = \begin{pmatrix} \delta & 0 & 0 \\ \delta A + \delta & \delta & 0 \\ \delta A^2 + \delta A + \delta & \delta A + \delta & \delta \end{pmatrix}$ where $\delta = B = \Delta t / z_r$. Let $A \approx 1$, $\delta = 0.001$:

$\mathcal{B} = \delta \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 3 & 2 & 1 \end{pmatrix}$. Stage cost: $\|\mathbf{X} - \theta^* \mathbf{1}\|^2 + 0.1 \|\mathbf{U}\|^2 = \|\mathcal{B}\mathbf{U} + (\mathcal{A}\theta^0 + \mathcal{D}) - \theta^* \mathbf{1}\|^2 + 0.1 \|\mathbf{U}\|^2$. Expanding: $J = \mathbf{U}^T (\mathcal{B}^T \mathcal{B} + 0.1 I_3) \mathbf{U} + 2(\mathcal{A}\theta^0 + \mathcal{D} - \theta^* \mathbf{1})^T \mathcal{B} \mathbf{U} + \text{const.}$
 So $Q_{QP} = 2(\mathcal{B}^T \mathcal{B} + 0.1 I_3)$ and $c_{QP} = 2\mathcal{B}^T (\mathcal{A}\theta^0 + \mathcal{D} - \theta^* \mathbf{1})$. Constraints: $\mathbf{U} \geq \mathbf{0}$ and $\mathbf{U} \leq 2\mathbf{1}$ (box constraints). This is directly passed to `cvxpy`.

Chapter 5

Machine Learning for Irrigation Intelligence

5.1 Overview: Four ML Tasks

The ML layer of Smart Agri-Togo performs four tasks:

1. **ET₀ forecasting** — predicting the 48-h disturbance sequence for MPC.
2. **Soil moisture nowcasting** — filling sensor gaps.
3. **Crop stress detection** — from camera images.
4. **Yield prediction** — mid-season forecast.

5.2 Supervised Learning Fundamentals

Given a dataset $\{(\mathbf{x}_i, y_i)\}_{i=1}^n$ of inputs and outputs, learn a function $\hat{f} : \mathbb{R}^d \rightarrow \mathbb{R}$ by minimising a **loss function**:

$$\hat{f} = \arg \min_{f \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^n \ell(f(\mathbf{x}_i), y_i) + \lambda \Omega(f), \quad (5.1)$$

where ℓ is the prediction error (e.g., squared error) and Ω is a regulariser.

Linear regression: $\hat{f}(\mathbf{x}) = \mathbf{w}^T \mathbf{x} + b$, minimising mean squared error (MSE): $\frac{1}{n} \sum_i (y_i - \mathbf{w}^T \mathbf{x}_i - b)^2$. Closed-form solution: $\mathbf{w} = (X^T X)^{-1} X^T \mathbf{y}$.

5.3 Long Short-Term Memory (LSTM) for ET_0 Forecasting

Evapotranspiration is a **time series**: today's ET_0 depends on the past few days of weather. An LSTM is a recurrent neural network that learns temporal dependencies.

At each time step t , the LSTM computes:

$$f_t = \sigma(W_f x_t + U_f h_{t-1} + b_f) \quad (\text{forget gate}) \quad (5.2)$$

$$i_t = \sigma(W_i x_t + U_i h_{t-1} + b_i) \quad (\text{input gate}) \quad (5.3)$$

$$\tilde{c}_t = \tanh(W_c x_t + U_c h_{t-1} + b_c) \quad (\text{candidate cell}) \quad (5.4)$$

$$c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t \quad (\text{cell state}) \quad (5.5)$$

$$o_t = \sigma(W_o x_t + U_o h_{t-1} + b_o) \quad (\text{output gate}) \quad (5.6)$$

$$h_t = o_t \odot \tanh(c_t) \quad (\text{hidden state}) \quad (5.7)$$

where σ is the sigmoid function, $\odot$ denotes element-wise multiplication. The LSTM **forgets** irrelevant past information (via f_t) and **remembers** new relevant information (via i_t).

Input: $\mathbf{x}_t = (T_a, RH, R_s, u_2)_t$ (past 72 h, sampled hourly = 288 features). **Output:** $\hat{ET}_0$ for next 48 steps.

5.4 Gaussian Process Regression for Soil Moisture

A **Gaussian Process** (GP) is a distribution over functions. Given observations $\{(\mathbf{x}_i, y_i)\}$, the GP provides a posterior mean $\mu(\mathbf{x}^*)$ and variance $\sigma^2(\mathbf{x}^*)$ at any test point $\mathbf{x}^*$.

The spatial covariance between cells i and j :

$$k(\mathbf{x}_i, \mathbf{x}_j) = \sigma_f^2 \exp\left(-\frac{\|\mathbf{x}_i - \mathbf{x}_j\|^2}{2\ell^2}\right), \quad (5.8)$$

where ℓ is the *length-scale* (how quickly correlation decays with distance). If cell (2, 3)'s sensor fails, the GP infers its moisture from cells (1, 3), (3, 3), (2, 2), (2, 4), weighted by their spatial correlation.

Exercise 5.5.1

You have 3 days of hourly ET_0 data: $\{ET_0^t\}_{t=1}^{72}$, plus weather API forecasts for temperature and humidity for the next 24 hours.

- (a) Describe the full input vector for the LSTM at time $t = 72$.

- (b) If the LSTM has 1 hidden layer with 64 units and predicts 48 future steps, how many trainable parameters does it have? (Count W , U , b for the gates and output layer.)
- (c) What loss function would you use for training?

Solution to Exercise 5.5.1

(a) Input $\mathbf{x}_{72}$: either (i) the full 72-step sequence of $(T_a, RH, R_s, u_2, ET_0) = 72 \times 5 = 360$ features, or (ii) (preferred for LSTM) feed the sequence step-by-step: at each of the 72 time steps, input 4 weather features + 1 $ET_0 = 5$ features, final state h_{72} carries the history. API forecasts for T_a and RH for next 24h are concatenated as static features (48 values).

(b) Input dim $d = 5$, hidden dim $h = 64$. Each of 4 gates has $W \in \mathbb{R}^{64 \times 5}$, $U \in \mathbb{R}^{64 \times 64}$, $b \in \mathbb{R}^{64}$: total per gate = $64 \times 5 + 64 \times 64 + 64 = 320 + 4096 + 64 = 4480$. For 4 gates: $4 \times 4480 = 17920$ LSTM params. Output layer: $\mathbb{R}^{64} \rightarrow \mathbb{R}^{48}$: $64 \times 48 + 48 = 3120$. Total: $\approx \mathbf{21\,040}$ parameters.

(c) Mean Squared Error (MSE): $\mathcal{L} = \frac{1}{48} \sum_{j=1}^{48} (\hat{ET}_0^j - ET_0^j)^2$. Alternatively, Mean Absolute Error (MAE) is more robust to outliers and preferred for physical quantities.

Part II

Irrigation Science and Modelling

Chapter 6

Soil Physics and the Water Balance

6.1 Soil Water Concepts

Definition: Key Soil Water Constants

- **Saturated water content** θ_s [m^3/m^3]: all pores filled. For sandy loam: ≈ 0.45 .
- **Field capacity** θ_{fc} : water retained after drainage ceases (~ 48 h). Sandy loam: ≈ 0.25 ; clay loam: ≈ 0.38 .
- **Permanent wilting point** θ_{wp} : plant cannot extract more water. Sandy loam: ≈ 0.10 ; clay loam: ≈ 0.22 .
- **Available water** $\theta_{aw} = \theta_{fc} - \theta_{wp}$ [m^3/m^3].
- **Readily available water** $\theta_{raw} = p\theta_{aw}$ where p is the depletion fraction before stress ($p = 0.30\text{--}0.50$, crop-dependent).

Example for southern Togo: Lomé soils are typically sandy loam to loam (FAO texture class). Using $\theta_{fc} = 0.30$, $\theta_{wp} = 0.12$, $\theta_s = 0.45$, $\theta_{aw} = 0.18 \text{ m}^3/\text{m}^3$.

For a root depth $z_r = 0.3$ m: total available water = $0.18 \times 0.3 \times 1000 = 54$ mm.

6.2 The van Genuchten Model

The **van Genuchten-Mualem** model relates pressure head h [m], water content θ , and hydraulic conductivity K :

$$\Theta = \frac{\theta - \theta_r}{\theta_s - \theta_r} = \frac{1}{[1 + |\alpha h|^n]^m}, \quad m = 1 - 1/n, \quad (6.1)$$

$$K(\Theta) = K_s \Theta^{1/2} [1 - (1 - \Theta^{1/m})^m]^2, \quad (6.2)$$

where θ_r is residual water content, α [m⁻¹] and $n > 1$ are shape parameters, K_s [m/h] is saturated conductivity. These parameters are fitted from laboratory measurements.

6.3 The Water Stress Coefficient K_s

When θ falls below the readily-available threshold $\theta_p = \theta_{fc} - p(\theta_{fc} - \theta_{wp})$:

$$K_s = \begin{cases} 1 & \text{if } \theta \geq \theta_p \\ \frac{\theta - \theta_{wp}}{\theta_p - \theta_{wp}} & \text{if } \theta_{wp} < \theta < \theta_p \\ 0 & \text{if } \theta \leq \theta_{wp} \end{cases} \quad (6.3)$$

K_s enters the ET calculation and the yield model directly.

Exercise 6.6.1

A clay loam soil in the field has $\theta_{fc} = 0.38$, $\theta_{wp} = 0.22$, $p = 0.45$ (onion).

- Compute the threshold θ_p and the readily available water depth for $z_r = 0.35$ m.
- If $\theta = 0.29$ m³/m³, compute K_s .
- If $K_s = 0.7$, by what factor does evapotranspiration reduce relative to the well-watered case?

Solution to Exercise 6.6.1

- $\theta_p = 0.38 - 0.45 \times (0.38 - 0.22) = 0.38 - 0.072 = 0.308$. Readily available water depth $= (\theta_{fc} - \theta_p) \times z_r = (0.38 - 0.308) \times 0.35 \times 1000 = 25.2$ mm.
- $K_s = \frac{0.29 - 0.22}{0.308 - 0.22} = \frac{0.07}{0.088} = 0.795$.
- $ET_c = K_c \cdot K_s \cdot ET_0$. With $K_s = 0.7$: $ET_c = 0.7 \cdot K_c \cdot ET_0$, i.e., 30% reduction relative to the non-stressed $ET_c = K_c \cdot ET_0$.

Chapter 7

Evapotranspiration — The Penman-Monteith Equation

7.1 What is Evapotranspiration?

Evapotranspiration (ET) is the water lost to the atmosphere via two processes: **evaporation** from the soil surface and **transpiration** from plant leaves. It is the main water demand term in the soil balance — managing ET is the central task of intelligent irrigation.

7.2 The FAO-56 Penman-Monteith Equation

The reference evapotranspiration ET_0 [mm/day] is computed as:

$$ET_0 = \frac{0.408 \Delta (R_n - G) + \gamma \frac{900}{T + 273} u_2 (e_s - e_a)}{\Delta + \gamma (1 + 0.34 u_2)} \quad (7.1)$$

Variables and how to compute them:

Δ [kPa/°C] — slope of saturation vapour pressure curve:

$$\Delta = \frac{4098 \left[0.6108 \exp\left(\frac{17.27T}{T+237.3}\right) \right]}{(T + 237.3)^2}. \quad (7.2)$$

e_s [kPa] — saturation vapour pressure:

$$e_s = \frac{e^\circ(T_{max}) + e^\circ(T_{min})}{2}, \quad e^\circ(T) = 0.6108 \exp\left(\frac{17.27T}{T + 237.3}\right). \quad (7.3)$$

e_a [kPa] — actual vapour pressure: $e_a = e_s \cdot RH/100$.

γ [kPa/°C] — psychrometric constant: $\gamma = 0.665 \times 10^{-3} P_{atm}$, where $P_{atm} \approx 101.3$ kPa at sea level.

R_n [MJ/m²/day] — net radiation = net shortwave minus net longwave: $R_n = (1 - \alpha)R_s - R_{nl}$, where $\alpha = 0.23$ (grass albedo) and R_{nl} is computed from temperature and actual sunshine hours.

G [MJ/m²/day] — soil heat flux: for daily calculations, $G \approx 0$.

u_2 [m/s] — wind speed at 2 m height (convert from height z : $u_2 = u_z \ln(2/0.000123)/\ln(z/0.000123)$).

Example — ET_0 computation for Lomé, 15 January

Data: $T_{max} = 34^\circ\text{C}$, $T_{min} = 24^\circ\text{C}$, $T_{mean} = 29^\circ\text{C}$, $RH_{mean} = 62\%$, $u_2 = 2.1$ m/s, $R_s = 18.4$ MJ/m²/day, latitude 6.14°N .

$e^\circ(34) = 0.6108 \times e^{17.27 \times 34 / (34 + 237.3)} = 0.6108 \times e^{2.098} = 0.6108 \times 8.152 = 4.978$ kPa. $e^\circ(24) = 2.984$ kPa. $e_s = \frac{4.978 + 2.984}{2} = 3.981$ kPa. $e_a = 3.981 \times 0.62 = 2.468$ kPa. $VPD = e_s - e_a = 1.513$ kPa. $\Delta = 4098 \times 3.267 / (266.3)^2 = 13384.5 / 70895.7 = 0.1888$ kPa/°C. $\gamma = 0.665 \times 10^{-3} \times 101.3 = 0.0674$ kPa/°C. $R_n \approx (1 - 0.23) \times 18.4 - 3.5 = 14.17 - 3.5 = 10.67$ MJ/m²/d.

Numerator: $0.408 \times 0.1888 \times 10.67 + 0.0674 \times \frac{900}{302} \times 2.1 \times 1.513 = 0.821 + 0.0674 \times 2.980 \times 2.1 \times 1.513 = 0.821 + 0.641 = 1.462$. Denominator: $0.1888 + 0.0674 \times (1 + 0.34 \times 2.1) = 0.1888 + 0.0674 \times 1.714 = 0.1888 + 0.1155 = 0.3043$. $ET_0 = 1.462 / 0.3043 = 4.80$ mm/day.

Exercise 7.7.1

During the hot dry period (March, Lomé): $T_{max} = 38^\circ\text{C}$, $T_{min} = 27^\circ\text{C}$, $RH = 48\%$, $u_2 = 3.0$ m/s, $R_s = 22$ MJ/m²/day.

- Compute ET_0 using the Penman-Monteith equation.
- If the crop coefficient for lettuce at mid-season is $K_c = 1.0$ and $K_s = 0.9$: what is the daily water demand per m² for a lettuce row?
- For a full row of 5 cells (25 m²), what is the total irrigation volume required per day (assuming no rain)?

Solution to Exercise 7.7.1

(a) $T_{mean} = 32.5^\circ\text{C}$. $e^\circ(38) = 0.6108 \times e^{1.886 \times 17.27 / (17.27 + 1)} \dots$ more carefully: $e^\circ(38) = 0.6108 \times \exp(17.27 \times 38 / 275.3) = 0.6108 \times \exp(2.382) = 0.6108 \times 10.83 = 6.615$ kPa. $e^\circ(27) = 3.601$ kPa. $e_s = 5.108$ kPa. $e_a = 5.108 \times 0.48 = 2.452$

kPa. VPD= 2.656 kPa. $\Delta = 4098 \times 4.031 / (269.8)^2 = 16518.7 / 72792 = 0.2269$ kPa/°C. $R_n \approx 0.77 \times 22 - 4.0 = 16.94 - 4.0 = 12.94$ MJ/m²/d. Num: $0.408 \times 0.2269 \times 12.94 + 0.0674 \times (900/305.5) \times 3.0 \times 2.656 = 1.198 + 1.770 = 2.968$. Den: $0.2269 + 0.0674 \times (1 + 1.02) = 0.2269 + 0.1361 = 0.363$. $ET_0 = 2.968 / 0.363 = \mathbf{8.18}$ mm/day.

(b) $ET_c = K_c \cdot K_s \cdot ET_0 = 1.0 \times 0.9 \times 8.18 = 7.36$ mm/day per m². In volume: 7.36×10^{-3} m/day $\times 1$ m² = 7.36 L/m²/day.

(c) For 25 m² (one full row): $7.36 \times 25 = \mathbf{184}$ L/day. Plus efficiency factor for drip (efficiency $\approx 90\%$): $184 / 0.9 \approx \mathbf{204}$ L of water to pump per row per day.

Chapter 8

Crop Water Requirements and Yield Models

8.1 Crop Coefficients (FAO-56)

The FAO-56 methodology separates the crop-specific water demand into three stages with tabulated coefficients:

Table 8.1: Crop coefficient K_c values (FAO-56) for our four crops

Crop	Cycle (d)	Init	Dev	Mid	Late	p
Onion	100	0.70	0.70–1.05	1.05	0.75	0.30
Carrot	90	0.70	0.70–1.05	1.05	0.95	0.35
Lettuce	55	0.70	0.70–1.00	1.00	0.95	0.30
Maize	110	0.30	0.30–1.20	1.20	0.35	0.55

8.2 FAO Yield Response Function

The relationship between water stress and yield loss (Stewart model):

$$\frac{Y_a}{Y_m} = \prod_{j \in \text{stages}} \left[1 - K_{y,j} \left(1 - \frac{ET_{a,j}}{ET_{m,j}} \right) \right], \quad (8.1)$$

where $K_{y,j}$ is the yield response factor (sensitivity of yield to stress in stage j). Table 8.2 gives FAO values.

Table 8.2: Yield response factors K_y by growth stage

Crop	Init	Dev	Mid	Late
Onion	0.45	0.80	0.30	0.20
Carrot	0.30	0.60	0.70	0.35
Lettuce	0.60	0.80	1.00	0.60
Maize	0.40	0.40	1.50	0.50

Exercise 8.8.1

An onion crop experiences 20% water deficit ($ET_a/ET_m = 0.80$) during the development stage only (all other stages fully irrigated). Maximum yield $Y_m = 20$ t/ha.

- Compute the actual yield Y_a using Eq. (8.1).
- Compute the revenue loss per hectare if the market price is 450 XOF/kg.
- What does this tell you about the economic value of the MPC over a hand-watered system that sometimes forgets to irrigate?

Solution to Exercise 8.8.1

- Only the development stage has stress: $K_{y,dev} = 0.80$, $ET_a/ET_m = 0.80$. $\frac{Y_a}{Y_m} = [1 - 0.80 \times (1 - 0.80)] \times 1 \times 1 \times 1 = 1 - 0.16 = 0.84$. $Y_a = 0.84 \times 20 = \mathbf{16.8}$ t/ha.
- Yield loss: $20 - 16.8 = 3.2$ t/ha = 3200 kg/ha. Revenue loss = $3200 \times 450 = \mathbf{1\,440\,000}$ XOF/ha.
- A single forgetful irrigation event causing 20% deficit during the sensitive development stage costs 1.44 million XOF/ha in lost revenue — more than the entire cost of the MPC hardware. The MPC pays for itself in one season by preventing a single stress event.

Part III

Organic Fertilization — No Chemicals, Only Life

Chapter 9

Soil Biology and Nutrient Cycles

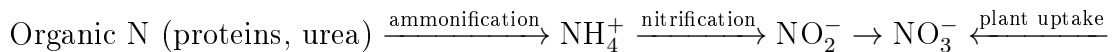
9.1 Why Organic? The Science of Soil Health

Synthetic fertilisers provide nutrients in ionic form (NO_3^- , NH_4^+ , H_2PO_4^-) that plants absorb immediately, but they do not build soil structure, they damage the soil microbiome over time, they leach into groundwater, and they require energy-intensive industrial production. For Smart Agri-Togo, the choice to use only organic inputs is:

- **Economically rational:** all inputs are locally available at near-zero cost.
- **Agronomically superior** in the long run: organic matter improves soil water-holding capacity, reducing ET and improving drought resistance.
- **Market-positioning:** “produits bio” can command 20–40% price premiums in Lomé’s upscale markets and export channels.

9.2 The Nitrogen Cycle

Nitrogen is the most important plant nutrient. The organic nitrogen cycle:



Bacteria (*Nitrosomonas*, *Nitrobacter*) drive nitrification. Temperature (25–35°C), moisture (50–60% water-filled pore space), and pH (6.5–7.5) are the controlling factors.

9.3 Nutrient Content of Organic Materials

Table 9.1: Typical nutrient content of organic inputs (% dry weight)

Material	N (%)	P (%)	K (%)	C:N ratio
Fresh cow manure	0.45	0.20	0.35	15–20
Composted cow manure	1.5	0.8	1.2	10–15
Fresh goat manure	0.75	0.35	0.65	12–16
Poultry manure (dry)	3.5	2.2	1.5	5–8
Human urine	4.0–8.0	0.5–1.5	1.0–3.0	<5
Composted feces	1.5	2.0	0.5	10–15
Biogas digestate (slurry)	2.0	0.8	1.5	8–12
Crop residues (maize stover)	0.6	0.1	1.2	50–80
Grass/green biomass	2.5	0.5	2.0	10–25
Wood ash	0	0.5	4.0–8.0	—

9.4 Nutrient Requirements of our Crops

Table 9.2: Total nutrient uptake per tonne of fresh yield (kg/t)

Crop	N (kg/t)	P (kg/t)	K (kg/t)	Target yield (t/ha)
Onion	2.5	0.5	2.0	20
Carrot	2.0	0.5	3.0	18
Lettuce	3.0	0.5	4.0	25
Maize	20	3.5	16	5

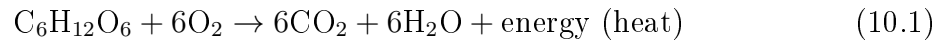
Total N requirement per hectare, e.g., for onion at 20 t/ha: $20 \times 2.5 = 50$ kg N/ha. Of this, perhaps 50% comes from soil mineralisation; we need to supply ≈ 25 kg N/ha from organic inputs.

Chapter 10

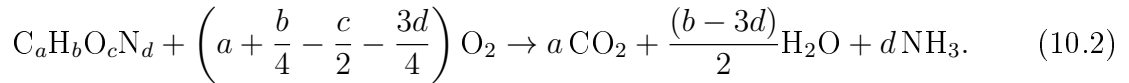
Composting: Theory and Practice

10.1 Biochemistry of Composting

Aerobic decomposition of organic matter by microbial communities:



More generally, for a substrate of composition $\text{C}_a\text{H}_b\text{O}_c\text{N}_d$:



10.1.1 The C:N Ratio

The optimal C:N ratio for rapid composting is **25–30:1**. Too high (C:N > 40): nitrogen deficiency slows decomposition. Too low (C:N < 15): ammonia volatilises and N is lost.

In practice, we blend:

- **Brown materials** (high C, low N): maize stover (C:N≈70), dry leaves (C:N≈60), rice straw (C:N≈80).
- **Green materials** (low C, high N): fresh grass (C:N≈15), food scraps (C:N≈15), animal manure (C:N≈12–20).

Blending formula: to achieve target C:N = r_T from two materials with C:N = r_1 and r_2 (with $r_1 > r_T > r_2$), the mass ratio $m_1 : m_2$ is:

$$\frac{m_1}{m_2} = \frac{N_2(r_T - r_2)}{N_1(r_1 - r_T)}, \quad (10.3)$$

where N_1, N_2 are the N contents (as fractions) of each material.

10.2 Composting Process Phases

1. **Mesophilic phase** (25–45°C, days 1–3): fast-growing bacteria break down easy carbohydrates.
2. **Thermophilic phase** (45–70°C, days 3–14): heat-tolerant bacteria destroy pathogens and weed seeds. **This is the key sanitisation step.**
3. **Cooling and mesophilic phase 2** (weeks 3–6): temperature drops, actinomyces and fungi break down lignin.
4. **Maturation / curing** (weeks 6–12): stable humic substances form.

10.3 Compost Pile Design

For a pile to self-heat, it must be at least 1 m × 1 m × 1 m. Optimal pile: 1.5 × 1.5 × 1.5 m. Moisture: 50–60% (squeeze test: a handful should just release a few drops). Aeration: turn the pile every 5–7 days.

Example — Compost blend for 1 ha onion field

Need: 30 t/ha compost (recommended dose for sandy loam in Togo). **Available:** maize stover (C:N=70, N=0.6%), goat manure (C:N=14, N=0.75%). **Target:** C:N=28.

Using Eq. (10.3): $m_1/m_2 = \frac{0.0075 \times (28-14)}{0.006 \times (70-28)} = \frac{0.105}{0.252} = 0.417$. So: 1 kg stover : 2.4 kg goat manure. For 30 tonnes of raw material (which will compost down to ~10 t mature compost after volume reduction), blend ≈ 9 t stover + 21 t goat manure.

Exercise 9.10.1

You have access to the following materials: 500 kg fresh grass (C:N=18, N=2.5%), 800 kg dry leaves (C:N=55, N=1.0%), 300 kg poultry manure (C:N=7, N=3.5%).

- (a) Compute the C:N ratio of the mixture.
- (b) Is this mixture suitable for composting without amendment? Justify.
- (c) If you want to raise the C:N to 28, how much additional dry leaves do you need to add?

Solution to Exercise 9.10.1

(a) Compute total C and N in the mixture. For a material with C:N ratio r and N fraction N_f (as decimal): $N_{\text{total}} = m \times N_f$; $C_{\text{total}} = m \times N_f \times r$. Grass: $N = 500 \times 0.025 = 12.5$ kg, $C = 12.5 \times 18 = 225$ kg. Dry leaves: $N = 800 \times 0.01 = 8$

kg, $C = 8 \times 55 = 440$ kg. Poultry: $N = 300 \times 0.035 = 10.5$ kg, $C = 10.5 \times 7 = 73.5$ kg. Total: $N_{tot} = 31$ kg, $C_{tot} = 738.5$ kg. C:N = $738.5/31 = \mathbf{23.8}$.

(b) C:N=23.8 is within the acceptable range (20–35) for composting. It is slightly below the optimal 28 but workable. The poultry manure provides abundant N for fast microbial activity.

(c) Add x kg dry leaves (C:N=55, $N_f = 0.01$): new totals: $N' = 31 + 0.01x$, $C' = 738.5 + 0.55x$. Target: $C'/N' = 28$. $738.5 + 0.55x = 28(31 + 0.01x) \Rightarrow 738.5 + 0.55x = 868 + 0.28x \Rightarrow 0.27x = 129.5 \Rightarrow x = \mathbf{480}$ kg additional dry leaves.

Chapter 11

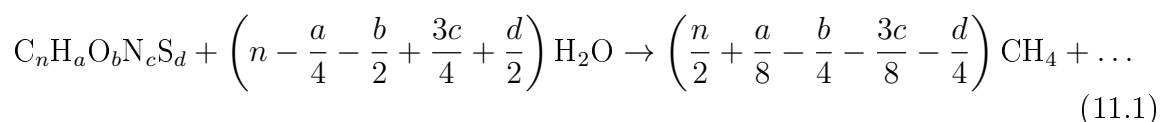
Biogas Digesters and Liquid Fertilizer

11.1 Anaerobic Digestion

Anaerobic digestion (AD) converts organic matter into **biogas** (energy) and **digestate** (fertiliser) in the absence of oxygen, through four stages:

1. **Hydrolysis**: complex polymers (carbohydrates, proteins, fats) $\rightarrow$ monomers (sugars, amino acids, fatty acids).
2. **Acidogenesis**: monomers $\rightarrow$ volatile fatty acids (VFAs), H_2 , CO_2 .
3. **Acetogenesis**: VFAs $\rightarrow$ acetate, H_2 , CO_2 .
4. **Methanogenesis**: $\text{CH}_3\text{COO}^- + \text{H}_2\text{O} \rightarrow \text{CH}_4 + \text{HCO}_3^-$ (acetoclastic) and $\text{CO}_2 + 4\text{H}_2 \rightarrow \text{CH}_4 + 2\text{H}_2\text{O}$ (hydrogenotrophic).

Overall stoichiometry (Buswell equation):



Practical biogas yields:

- Cattle manure: 22–28 L biogas/kg fresh weight, 60% CH_4 .
- Food waste: 80–100 L biogas/kg.
- Human feces: 30–40 L biogas/kg.
- Mixed farm waste (our case): ~ 30 L biogas/kg fresh input.

11.2 Fixed-Dome Biogas Digester Design

The fixed-dome (Deenbandhu or Chinese model) is the most appropriate for Togo: low cost, no moving parts, locally constructable, long lifespan (20+ years).

Design equations:

$$V_{\text{digester}} = Q_{\text{input}} \times HRT, \quad (11.2)$$

where Q_{input} [m³/day] is the daily slurry volume and HRT is the Hydraulic Retention Time (typically 30–40 days at 30–35°C in Togo's climate).

For a farm producing D kg/day of manure:

$$Q_{\text{input}} = \frac{D}{d_s} \text{ m}^3/\text{day}, \quad (11.3)$$

where $d_s \approx 1000\text{--}1050$ kg/m³ is the slurry density (1:1 manure:water dilution).

Biogas production: $Q_{\text{gas}} = Y_{\text{gas}} \times D$ [L/day], where Y_{gas} is the specific yield.

Digestate use: The liquid digestate is a highly available N, P, K source. It can be diluted 1:5 with water and applied through the drip irrigation system (**fertigation**) — a key advantage of the drip system.

Example — Biogas digester for Smart Agri-Togo

Available: 2 cows (12 kg dung/cow/day), 10 goats (1.5 kg/goat/day), kitchen waste from on-site shelter (3 kg/day). Total daily input: $D = 24 + 15 + 3 = 42$ kg/day. $Q_{\text{input}} = 42/1000 = 0.042$ m³/day (after 1:1 dilution, actually 0.084 m³/day). $V_{\text{digester}} = 0.084 \times 35 = 2.94$ m³. Use a **3 m³** fixed-dome digester. Biogas: $42 \times 30 = 1260$ L/day = 1.26 m³/day. Energy: 1 m³ CH₄ $\approx$ 35.9 MJ. With 60% CH₄: $1.26 \times 0.6 \times 35.9 = 27.1$ MJ/day $\approx$ cooking energy for 5–6 people. This **replaces LPG or charcoal** for the on-site shelter. Digestate: $\approx$ 30 kg/day of nutrient-rich slurry available for fertigation.

Chapter 12

Human Biosolids — Safe, Scientific, Sovereign

12.1 The Taboo That Costs Farmers Billions

Human urine and faeces contain the exact same plant nutrients as expensive synthetic fertilisers — because they *are* the end products of digesting those nutrients. Recovering them closes the nutrient cycle, reduces pollution, and saves money. This practice has been used for millennia and is endorsed by the WHO, FAO, and UNEP when done safely.

Nutrient Content of Human Excreta

- **Urine:** ~ 2 g N/L, 0.25 g P/L, 1.5 g K/L. A person produces ~ 1.5 L/day, providing ~ 3 g N/day. One person's annual urine supply = ~ 1.1 kg N + 0.14 kg P + 0.8 kg K — equivalent to 5 kg of urea fertiliser. Urine is sterile (in a healthy person) and safe to use diluted 1:10 on growing crops.
- **Faeces:** ~ 30 g dry weight/person/day, 5–8% N, 2–5% P. Requires treatment before safe use (see below).

12.2 Treatment Methods

12.2.1 Urine (Struvite / Direct Application)

Urine can be stored in a sealed container for 1 month (urea hydrolysis raises pH to ~ 9 , killing pathogens) then diluted 1:10 and applied to soil, not directly on leaves. Alternatively, **struvite** ($\text{MgNH}_4\text{PO}_4 \cdot 6\text{H}_2\text{O}$) can be precipitated:



Struvite is a slow-release N-P fertiliser. By adding magnesium chloride (MgCl_2) at a Mg:P molar ratio of 1.2:1 and adjusting pH to 8.5–9.5, 80–90% of P and 30–40% of N precipitate.

12.2.2 Faecal Matter: Composting and Vermicomposting

WHO category: faecal matter is a Category A biological hazard; it must be treated to reduce helminth eggs (<1 egg/g) and pathogen indicators ($<10^3$ faecal coliforms/g) before safe agricultural use.

Thermophilic composting: co-compost with bulking material (straw, wood chips) in a 2-chamber composter. Temperature must exceed 55°C for ≥ 3 days (WHO guideline). After 6 months of maturation, the product is safe for use on crops not eaten raw.

Vermicomposting: add earthworms (*Eisenia fetida*) after thermophilic phase. Worm castings (vermicompost) are exceptional soil conditioners, with $2\text{--}5\times$ higher nutrient availability than ordinary compost.

12.2.3 The Urine Diversion Dry Toilet (UDDT)

A UDDT separates urine from faeces at the source:

- **Urine diverter:** a specially shaped toilet bowl collects urine in a front channel, faeces in the rear.
- **Urine container:** 20-L HDPE jerricans, stored in a cool dark place for 1 month.
- **Dry vault:** faeces fall onto drying ash/lime material; no flush water required. Filled vault composted for 6 months.

Cost: a UDDT can be constructed locally for $\sim 50\,000\text{--}100\,000$ XOF. It requires no water for flushing, solving sanitation and fertiliser supply simultaneously.

12.3 WHO Safety Guidelines Summary

Table 12.1: WHO safety requirements for use of human excreta in agriculture

Product	Treatment requirement	Application restriction
Urine (stored 1 mo)	None (sterile)	Dilute 1:10, not on edible parts
Vermicompost	Thermophilic pre-treatment	Any crop, avoid 2 wk before harvest
Compost ($>55^\circ\text{C}$, 3d)	Pathogen reduction verified	Avoid crops eaten raw
Digestate (biogas)	AD process (35°C , 30d HRT)	Dilute, drip application preferred

Chapter 13

Integrated Organic Fertilization Plan

13.1 Annual Nutrient Budget for Smart Agri-Togo (1 ha)

Table 13.1: Annual N supply from all organic sources (1 ha)

Source	Quantity	N conc.	Avail. factor	N supplied (kg)
Composted cow manure	15 t/ha	1.5%	0.50	112.5
Biogas digestate	5 t/ha	2.0%	0.80	80.0
Stored urine (5 people)	2.7 t/ha	4.0%	0.90	97.2
Vermicompost (feces)	1 t/ha	2.5%	0.70	17.5
Wood ash	0.5 t/ha	0%	—	0.0
Total N				307.2 kg/ha
<i>Total N demand (all 4 crops)</i>				165 kg/ha
<i>N balance</i>				+142.2 (surplus — build soil c

13.2 Application Calendar

Table 13.2: Organic input application calendar

Timing	Input	Method
4 weeks before planting	Composted manure 15 t/ha	Incorporate to 20 cm depth
At planting	Vermicompost 1 t/ha	Band-placed near root zone
Week 3	Urine solution 1:10	Fertigation via drip (2 L/m ²)
Week 5	Biogas digestate 1:5	Fertigation via drip (2 L/m ²)
Week 7	Urine solution 1:10	Fertigation (top dressing)
Week 9	Biogas digestate 1:5	Fertigation
Post-harvest	Crop residues	Mulch or compost pile

Exercise 10.13.1

You have 5 workers on the farm, each producing 1.5 L urine/day.

- Compute the annual urine production and total N available after 1-month storage.
- How many litres of 1:10 diluted urine solution does this produce for fertigation?
- If drip application rate is 2 L/m² per application, how many 1-ha applications does this cover?

Solution to Exercise 10.13.1

- Annual production: $5 \times 1.5 \times 365 = 2737.5 \text{ L} \approx 2740 \text{ L/year}$. N content: $2 \times 10^{-3} \text{ kg N/L} \times 2740 = 5.48 \text{ kg N}$. With 90% availability: $5.48 \times 0.9 = 4.93 \text{ kg N}$. Small but valuable (supplement to compost).
- Diluted at 1:10: each litre of urine gives 11 L of solution. Total solution: $2740 \times 11 = 30140 \text{ L} = 30.1 \text{ m}^3$ of solution.
- For 1 ha = 10000 m² at 2 L/m²: need 20000 L = 20 m³ per application. Number of applications: $30.1/20 = 1.5$. So roughly **1–2 full-field applications per season** from 5 workers' urine. Scaled to 10 workers (include farm visitors/family): 3 applications.

Part IV

Energy Systems for Smart Agri-Togo

Chapter 14

Energy Load Analysis

14.1 The Energy Question

Every component of Smart Agri-Togo consumes electricity: the pump, the sensors, the Raspberry Pi, the lights in the field shelter, the camera, the mobile phone charger. We must first *quantify* this consumption precisely before designing any power system. The design principle: **minimise but never compromise**.

14.2 Fundamental Concepts

Definition: Power, Energy, and Peak Load

- **Power** P [W or kW]: rate of energy consumption at any instant.
- **Energy** E [Wh or kWh]: power integrated over time — $E = P \times t$.
- **Daily energy demand** E_d [kWh/day]: sum of all loads.
- **Peak load** P_{peak} [kW]: maximum instantaneous demand (determines inverter size).

14.3 Pump Power

The **hydraulic power** required to lift water from depth H_{static} [m] at flow rate Q [m³/s] is:

$$P_{hydraulic} = \rho g Q H_{total}, \quad (14.1)$$

where $\rho = 1000$ kg/m³, $g = 9.81$ m/s², $H_{total} = H_{static} + H_{friction} + H_{pressure}$.

The **electrical power drawn** by the pump motor:

$$P_{electric} = \frac{P_{hydraulic}}{\eta_{pump} \times \eta_{motor}}, \quad (14.2)$$

where $\eta_{\text{pump}} \approx 0.65\text{--}0.75$ (pump efficiency) and $\eta_{\text{motor}} \approx 0.85\text{--}0.92$.

Example — Pump power for Smart Agri-Togo

Data: $Q = 0.8 \text{ L/s} = 8 \times 10^{-4} \text{ m}^3/\text{s}$. $H_{\text{static}} = 15 \text{ m}$ (borehole depth to water table) + 5 m (tank height) = 20 m . $H_{\text{friction}} \approx 3 \text{ m}$ (pipe losses). $H_{\text{pressure}} \approx 5 \text{ m}$ (drip system operating pressure). $H_{\text{total}} = 28 \text{ m}$. $\eta_{\text{pump}} = 0.68$, $\eta_{\text{motor}} = 0.88$. $P_{\text{hydraulic}} = 1000 \times 9.81 \times 8 \times 10^{-4} \times 28 = 220 \text{ W}$. $P_{\text{electric}} = 220 / (0.68 \times 0.88) = 220 / 0.598 = \mathbf{368 \text{ W} \approx 0.37 \text{ kW}}$.

14.4 Complete Load Inventory

Table 14.1: Energy load inventory for Smart Agri-Togo

Equipment	Power (W)	Hours/day	Energy (Wh/day)	Notes
Submersible pump	370	5	1850	Avg over season
Raspberry Pi 4	6	24	144	Always on (control)
Arduino Mega $\times 4$	0.25	24	24	Low power mode
Soil sensors $\times 25$	0.5	24	300	Capacitive touch
Weather station	0.5	24	12	Ecowitt/Da
GSM/4G modem	2	24	48	SIM7600
Camera (plant health)	3	2	6	Daily scan
LED field lights	50	4	200	$5 \times 10 \text{ W LED}$
LED shelter/lab	60	6	360	Work space
Laptop (analysis)	45	4	180	Research comp
Phone charging $\times 3$	10	2	60	Farm work
Misc / safety margin (+20%)	—	—	637	
Total daily demand E_d			3821 Wh/day	$\approx 3.82 \text{ kWh/day}$
Peak load P_{peak}	$\approx 520 \text{ W}$		Pump + lights +	

Energy Optimisation Principles

1. Use **LED** lighting throughout ($10\times$ more efficient than incandescent).
2. Pump only during **peak solar hours** (10:00–15:00) to match generation.
3. Put the Raspberry Pi in **low-power mode** during non-control periods.
4. Use **capacitive** (not resistive) soil sensors — $100\times$ lower power.
5. Schedule the laptop for daytime (solar) use only.

Chapter 15

Solar Photovoltaic Fundamentals

15.1 The Photovoltaic Effect

A solar cell (typically monocrystalline silicon, efficiency 20–22%) converts photon energy into electrical current via the photoelectric effect. The current-voltage (I-V) characteristic:

$$I = I_{ph} - I_0 \left[\exp\left(\frac{q(V + IR_s)}{nk_B T}\right) - 1 \right] - \frac{V + IR_s}{R_{sh}}, \quad (15.1)$$

where I_{ph} is the photocurrent (proportional to irradiance), I_0 is the dark saturation current, $q = 1.6 \times 10^{-19}$ C, n is the ideality factor (≈ 1.3), $k_B = 1.38 \times 10^{-23}$ J/K, R_s is series resistance, and R_{sh} is shunt resistance.

Key points on the I-V curve:

- **Short-circuit current** I_{sc} : maximum current, at $V = 0$. Proportional to irradiance.
- **Open-circuit voltage** V_{oc} : maximum voltage, at $I = 0$. ≈ 0.6 – 0.65 V per cell.
- **Maximum power point** (MPP): the (V_{MPP}, I_{MPP}) pair maximising $P = VI$. An **MPPT controller** tracks this point continuously.
- **Fill Factor**: $FF = \frac{P_{MPP}}{V_{oc} \times I_{sc}} \approx 0.75$ – 0.82 for quality modules.

15.2 Solar Resource in Togo

Table 15.1: Solar irradiance data for Lomé, Togo (monthly averages)

Month	GHI (kWh/m ² /day)	Peak Sun Hours	Temperature effect
October	5.2	5.2	High T penalty 3%
November	5.6	5.6	Moderate
December	5.8	5.8	Best month
January	5.5	5.5	Harmattan dust (-5%)
February	5.4	5.4	Harmattan
March	5.8	5.8	High irradiance
April	5.3	5.3	Start of clouds
Season avg.	5.4	5.4	

Important: During the Harmattan season (December–February), dust particles can reduce panel output by 5–15%. Panels must be cleaned weekly during this period.

15.3 Why Solar is Optimal for Togo

Comparison of Power Sources for Smart Agri-Togo

- **Grid connection (CEET):** Cost of grid extension to a peri-urban field: 2–5 million XOF. Electricity tariff: ≈ 80 XOF/kWh for industry. Grid reliability: 60–70% uptime (frequent cuts). Control systems require 24/7 power — unacceptable.
- **Diesel generator:** Initial cost: 500 000 XOF (3 kVA). Fuel: ≈ 0.8 L diesel/kWh $\times 3.82$ kWh/day = 3.1 L/day $\times 365$ = 1130 L/year $\times 700$ XOF/L = 791 000 XOF/year fuel alone. Maintenance: high. CO₂ emissions: 3 kg/L $\times 1130$ = 3.4 t/year. Not compatible with organic/bio brand.
- **Solar PV + battery (recommended):** Initial cost ≈ 1.8 –2.5 million XOF. Annual operating cost: near zero (cleaning, battery replacement every 10 years). Zero fuel, zero emissions. 24/7 reliable power. **Payback vs diesel: 2–3 years.**
- **Hybrid solar + grid:** Possible if grid connection already exists. Solar covers daytime loads; grid as backup. Reduces dependence on either.

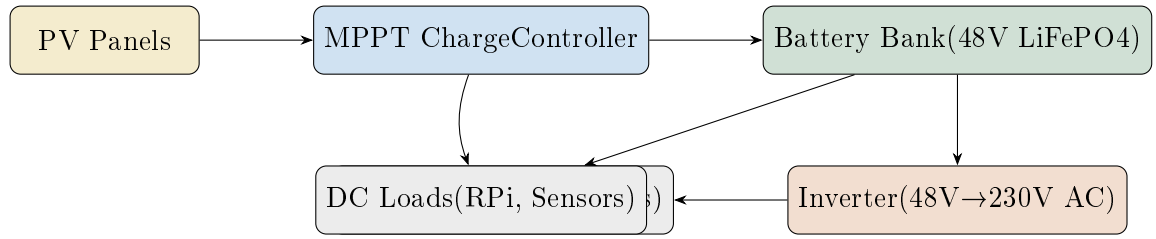
Verdict: Off-grid solar PV with battery storage is the optimal choice.

Chapter 16

Solar System Design for Smart Agri-Togo

16.1 System Architecture

The recommended system is a **48V DC bus** with:



16.2 Step-by-Step System Sizing

16.2.1 Step 1: Correct Daily Load for System Losses

$$E_{corrected} = \frac{E_d}{\eta_{system}} = \frac{3820}{0.80} = 4775 \text{ Wh/day} \approx 4.78 \text{ kWh/day}, \quad (16.1)$$

where $\eta_{system} = 0.80$ accounts for battery charging/discharging losses (0.92), inverter losses (0.93), and wiring losses (0.97): $\eta = 0.92 \times 0.93 \times 0.97 \approx 0.83 \approx 0.80$ (conservative).

16.2.2 Step 2: PV Array Sizing

$$P_{PV} = \frac{E_{corrected}}{PSH \times \eta_{temp}} = \frac{4780}{5.4 \times 0.94} = \frac{4780}{5.076} = 942 \text{ W}_p \approx 1.0 \text{ kW}_p, \quad (16.2)$$

where $\eta_{temp} = 0.94$ accounts for temperature derating at 35°C (panels lose $\approx 0.4\%/C$ above 25°C, so $10 \times 0.4 = 4\%$ loss).

Panel selection: Use 375 W_p monocrystalline panels (commercially available in Accra/Lomé). Number of panels: $1000/375 = 2.67 \Rightarrow$ **3 panels of 375 W_p = 1125 W_p** (+12.5% margin).

Array configuration: In a 48V system, the minimum open-circuit voltage is $V_{oc,min} \approx 55$ V. One panel: $V_{oc} \approx 46$ V, $V_{MPP} \approx 38$ V. Connect **2 panels in series:** $V_{oc,string} = 92$ V, $V_{MPP,string} = 76$ V (compatible with most 48V MPPT controllers). The third panel is on a separate string or connected in a 1S1P configuration with a second MPPT input.

16.2.3 Step 3: Battery Bank Sizing

Autonomy required: $N_{aut} = 2$ days (2 cloudy days without solar). Maximum depth of discharge: $DoD = 0.80$ (LiFePO4 chemistry recommended).

$$E_{bat} = \frac{E_d \times N_{aut}}{DoD} = \frac{3820 \times 2}{0.80} = 9550 \text{ Wh} \approx \mathbf{9.6 \text{ kWh}}. \quad (16.3)$$

In a 48V system: $C_{bat} = E_{bat}/V = 9600/48 = 200$ Ah.

Battery selection: 2×100 Ah LiFePO4 batteries at 48V (in parallel) = 200 Ah total. Alternatively, 1×200 Ah LiFePO4 48V battery pack. LiFePO4 is preferred over lead-acid for:

- 3000+ charge cycles at 80% DoD (~ 10 years lifespan) vs 500 cycles for AGM lead-acid.
- Temperature performance in Togo's heat (30–35°C is optimal for LiFePO4).
- Built-in Battery Management System (BMS).
- 50% lighter and more compact.

16.2.4 Step 4: MPPT Charge Controller

Maximum PV current (short-circuit): $I_{sc,total} = 3 \times 10 = 30$ A (for 3 panels with $I_{sc} = 10$ A each, 2 in series + 1 separate). Select a **60A MPPT controller** with 48V battery voltage and maximum PV input voltage ≥ 120 V (safety margin above $V_{oc,string} = 92$ V).

16.2.5 Step 5: Inverter

Peak load $P_{peak} = 520$ W. Surge capacity needed for pump startup ($3\text{--}5 \times$ nameplate): $520 \times 3 = 1560$ W. Select a **2 kVA (2000 W) pure sine wave inverter, 48V DC input, 230V AC output**. Pure sine wave is mandatory for the variable-speed pump and electronic loads.

16.3 Complete System Specification

Table 16.1: Final solar system specifications

Component	Specification	Estimated Cost (XOF)
PV modules	$3 \times 375 \text{ W}_p$ mono, 48V string	360 000
MPPT controller	Victron SmartSolar 75/60A	180 000
Battery bank	200 Ah / 48V LiFePO4	600 000
Inverter	2 kVA pure sine, 48V, Victron/Growatt	250 000
Mounting structure	Galvanised steel, 15° tilt	80 000
Wiring & protection	6 mm ² PV cable, circuit breakers	60 000
Monitoring system	Victron CCGX / Raspberry Pi addon	80 000
Total		≈1 610 000 XOF
<i>Annual savings vs diesel: 791 000 XOF. Payback: ≈ 2 years.</i>		

16.4 Panel Placement and Tilt Optimisation

In Togo (latitude $\phi \approx 6\text{N}$), the optimal panel tilt angle for year-round energy maximisation:

$$\beta_{opt} \approx |\phi| + 10 = 6 + 10 = 16 \approx \mathbf{15}. \quad (16.4)$$

Panels should face **south** (or slightly southeast to capture morning sun during harmattan when afternoons are dustier). The shelter/lab building can serve as the mounting base.

Exercise 11.16.1

During the harmattan (January), dust reduces panel output by 12%. The pump runs 6 hours/day instead of 5 (drier conditions).

- Recompute the new daily energy demand.
- Recompute the available solar generation with 3 panels of 375 W_p at 5.5 PSH and 12% derating.
- Is there an energy deficit? If so, by how much, and what strategy do you recommend?

Solution to Exercise 11.16.1

(a) Original pump energy: $370 \times 5 = 1850$ Wh. New pump energy: $370 \times 6 = 2220$ Wh. Difference: $+370$ Wh. New total: $3820 + 370 = 4190$ Wh/day. Corrected: $4190/0.80 = 5238$ Wh/day.

(b) Available generation: $1125 W_p \times 5.5 \text{ PSH} \times (1 - 0.12) \times 0.94 = 1125 \times 5.5 \times 0.88 \times 0.94 = 1125 \times 4.577 = 5149$ Wh/day.

(c) Deficit: $5238 - 5149 = 89$ Wh/day (small, $\approx 2\%$). Strategies:

- i. **Clean panels daily** during harmattan — restores most of the 12% loss.
- ii. **Reduce pump run time** by 15 min — saves ≈ 90 Wh.
- iii. Draw ≈ 90 Wh from battery reserve (negligible — battery holds 9600 Wh).

Practical recommendation: clean panels every 2 days during harmattan season (15 minutes of work), completely eliminating the deficit.

Exercise 12.16.2

System economics:

- (a) Compute the levelised cost of electricity (LCOE) for the solar system over 20 years, assuming: initial cost = 1 610 000 XOF, annual maintenance = 30 000 XOF, annual generation = $1125 \times 5.4 \times 365 \times 0.80 = 1780$ kWh (accounting for system efficiency).
- (b) Compare to: (i) grid electricity at 80 XOF/kWh, (ii) diesel at 700 XOF/L consuming 0.8 L/kWh.
- (c) At what year does solar break even vs diesel?

Solution to Exercise 12.16.2

(a) Total 20-year cost: $1\,610\,000 + 20 \times 30\,000 = 2\,210\,000$ XOF. Total energy: $1780 \times 20 = 35\,600$ kWh. LCOE = $2\,210\,000/35\,600 = \mathbf{62.1}$ XOF/kWh.

(b) Grid: 80 XOF/kWh. Diesel: $0.8 \times 700 = 560$ XOF/kWh. Solar LCOE (62 XOF/kWh) is cheaper than both grid (80) and diesel (560).

(c) Annual energy consumption: ≈ 1400 kWh (net useful energy). Annual diesel cost: $1400 \times 560 = 784\,000$ XOF/year. Solar annual cost equivalent: $2\,210\,000/20 + 1400 \times 0 = 110\,500 + 0 = 110\,500$ XOF/year. Savings vs diesel: $784\,000 - 110\,500 = 673\,500$ XOF/year. Payback: $1\,610\,000/673\,500 = \mathbf{2.4}$ years. After payback, nearly pure profit on energy savings.

Appendix A

Quick Reference: Key Equations

Equation	Formula	Chapter
Soil water balance	$z_r \dot{\theta} = I + P - ET_c - D$	1
Euler method	$\theta^{k+1} = \theta^k + \frac{\Delta t}{z_r} (I^k - ET_c^k)$	1
Lifted MPC system	$\mathbf{X} = \mathcal{A}\boldsymbol{\theta}^0 + \mathcal{B}\mathbf{U} + \mathcal{D}$	2
KKT stationarity	$\nabla f + \sum \mu_i \nabla g_i = \mathbf{0}$	3
MPC QP cost	$J = \ \boldsymbol{\theta} - \boldsymbol{\theta}^*\ _Q^2 + \ \mathbf{u}\ _R^2$	4
Penman-Monteith	$ET_0 = \frac{0.408\Delta(R_n - G) + \gamma \frac{900}{T+273} u_2 (e_s - e_a)}{\Delta + \gamma(1+0.34u_2)}$	7
Stress coefficient	$K_s = (\theta - \theta_{wp})/(\theta_p - \theta_{wp})$	6
Yield response	$Y_a/Y_m = \prod_j [1 - K_{y,j}(1 - ET_{a,j}/ET_{m,j})]$	8
Compost C:N blend	$m_1/m_2 = N_2(r_T - r_2)/[N_1(r_1 - r_T)]$	10
Digester volume	$V = Q_{input} \times HRT$	11
Hydraulic power	$P_{hyd} = \rho g Q H$	14
PV array size	$P_{PV} = E_{corrected}/(PSH \times \eta_{temp})$	16
Battery bank	$E_{bat} = E_d \times N_{aut}/DoD$	16

Appendix B

Numerical Values for Lomé, Togo

Parameter	Value
Latitude	6.14°N
Annual mean temperature	27°C
Mean daily solar radiation (Oct–Apr)	5.4 kWh/m ² /day (GHI)
Harmattan period	Dec–Feb (dust reduces solar by 5–15%)
Mean wind speed at 2 m	2.0–3.5 m/s
Mean relative humidity (dry season)	55–70%
Mean ET ₀ (dry season)	4.5–8.5 mm/day
Typical sandy loam θ_{fc}	0.28–0.32 m ³ /m ³
Typical sandy loam θ_{wp}	0.10–0.14 m ³ /m ³
CEET electricity tariff (industrial)	≈80 XOF/kWh
Diesel price (2024)	≈700 XOF/L
1 USD ≈	610 XOF

Appendix C

Python Code Snippets

C.1 Soil Water Balance Euler Integration

```
import numpy as np

z_r = 300 # root zone depth [mm]
dt = 0.5 # time step [h]
theta = np.zeros(49) # 24h / 0.5h = 48 steps + initial
theta[0] = 0.55 # initial moisture

ET_c = 0.30 # [mm/h], assumed constant
I = np.zeros(48)
I[12:18] = 0.8 # irrigation 6h-9h (steps 12-17)

for k in range(48):
    theta[k+1] = theta[k] + dt/z_r * (I[k] - ET_c)
    theta[k+1] = max(0.12, min(0.82, theta[k+1])) # clamp

print(f"Final moisture: {theta[-1]:.3f}")
```

C.2 Penman-Monteith ET_0 Calculation

```
import numpy as np

def ET0_PM(T_max, T_min, RH_mean, u2, Rs, lat_deg):
    T = (T_max + T_min) / 2
    # Saturation vapour pressure
    es_max = 0.6108 * np.exp(17.27*T_max/(T_max+237.3))
```

```
es_min = 0.6108 * np.exp(17.27*T_min/(T_min+237.3))
es = (es_max + es_min) / 2
ea = es * RH_mean / 100
# Slope of saturation vapour pressure curve
delta = 4098 * (0.6108*np.exp(17.27*T/(T+237.3))) / (T+237.3)**2
gamma = 0.0674 # psychrometric constant [kPa/degC] at sea level
# Net radiation (simplified)
Rn = 0.77 * Rs - 3.5
G = 0.0
# Penman-Monteith
num = 0.408*delta*(Rn-G) + gamma*(900/(T+273))*u2*(es-ea)
den = delta + gamma*(1 + 0.34*u2)
return num / den

ET0 = ET0_PM(T_max=34, T_min=24, RH_mean=62, u2=2.1, Rs=18.4, lat_deg=6.14)
print(f"ET0 = {ET0:.2f} mm/day")
```

C.3 Simple MPC Solver (cvxpy)

```
import cvxpy as cp
import numpy as np

N = 5 # cells (one row)
H = 48 # horizon (24h at 30-min steps)
dt, zr = 0.5, 300.0
delta = dt / zr
alpha, beta = 1.0, 0.1
theta_star = 0.65 * np.ones(N)
I_max = 2.0 # mm/h
Q_max = 5.0 # total flow mm/h

theta0 = np.array([0.42, 0.55, 0.48, 0.60, 0.38])
ET_c_forecast = 0.30 * np.ones((H, N)) # from LSTM

U = cp.Variable((H, N))
theta = {0: theta0}
cost = 0
constraints = []
```

```
for k in range(H):
    theta[k+1] = theta[k] + delta*(U[k] - ET_c_forecast[k])
    cost += alpha * cp.sum_squares(theta[k+1] - theta_star)
    cost += beta * cp.sum_squares(U[k])
    constraints += [U[k] >= 0, U[k] <= I_max]
    constraints += [cp.sum(U[k]) <= Q_max]

prob = cp.Problem(cp.Minimize(cost), constraints)
prob.solve(solver=cp.OSQP)
print("Optimal first irrigation:", U.value[0])
```

Appendix D

Glossary

Term	Meaning
θ [m ³ /m ³]	Volumetric soil water content
θ_{fc}	Field capacity
θ_{wp}	Permanent wilting point
θ^*	Optimal irrigation target
K_c	Crop coefficient (FAO-56)
K_s	Water stress coefficient
K_y	Yield response factor
ET_0	Reference evapotranspiration (Penman-Monteith)
ET_c	Crop evapotranspiration ($= K_c K_s ET_0$)
z_r [m]	Root zone depth
H (MPC)	Prediction horizon (number of steps)
J	Optimal control cost functional
α, β	Cost weighting parameters (yield vs water)
Q_{max}	Maximum pump flow rate (mm/h)
R_n	Net radiation [MJ/m ² /day]
Δ	Slope of saturation vapour pressure curve [kPa/°C]
γ	Psychrometric constant [kPa/°C]
e_s, e_a	Saturation and actual vapour pressure [kPa]
PSH	Peak Sun Hours [h/day]
LCOE	Levelised Cost of Electricity [XOF/kWh]
DoD	Depth of Discharge (battery)
C:N	Carbon to Nitrogen ratio (composting)
AD	Anaerobic Digestion
UDDT	Urine Diversion Dry Toilet
HRT	Hydraulic Retention Time (digester)

MPC	Model Predictive Control
LSTM	Long Short-Term Memory (neural network)
GPR	Gaussian Process Regression
QP	Quadratic Program
KKT	Karush-Kuhn-Tucker (optimality conditions)
PMP	Pontryagin's Maximum Principle
MPPT	Maximum Power Point Tracking (solar)
